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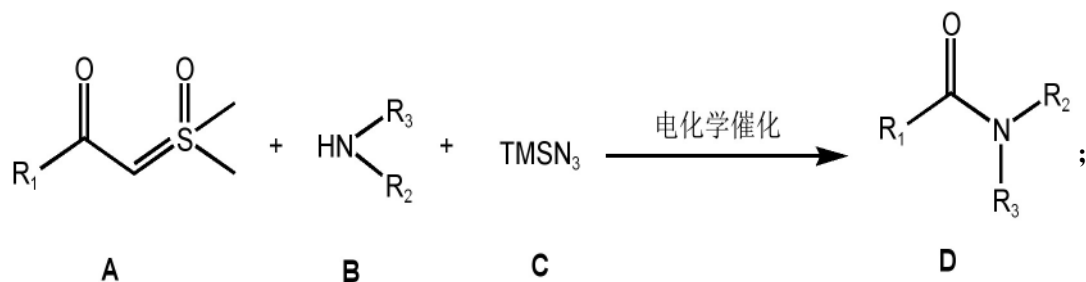
(54) 发明名称

一种电化学合成酰胺类化合物的方法

(57) 摘要

本发明提出了一种电化学合成酰胺类化合物的新方法,该方法以 α -羰基硫叶立德、烷基胺和叠氮基三甲基硅烷为原料,通过电化学反应高效构建酰胺结构,能以近乎当量收率得到目标酰胺化产物。本发明方法在温和条件下进行,无需缩合剂和催化剂,无额外的副产物,展现了环境友好性;具有良好的官能团耐受性,反应收率极高且纯化方式简单,并能应用于电化学连续流反应,适合工业化生产,具有极高产业化应用价值。

1. 一种电化学合成酰胺类化合物D的方法,以化合物A,B,C为原料,反应式如下:



其中,

R_1 选自 C_1-6 的烷基或环烷基、 R_4 -, 或 , X为O或S;

R_2 选自 C_1-6 的烷基或环烷基、 R_5 -, 或 , X为O或S;m为1-3的整数;

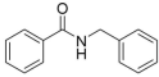
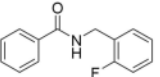
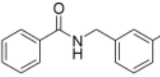
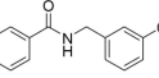
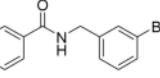
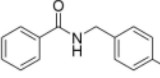
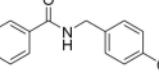
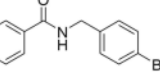
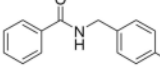
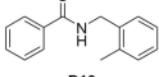
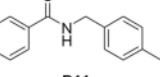
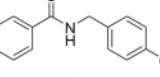
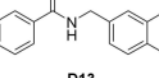
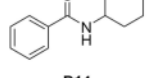
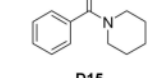
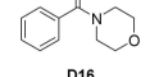
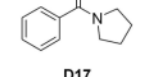
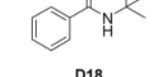
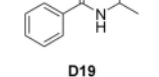
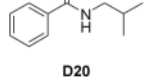
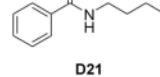
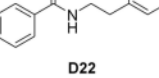
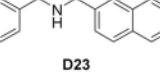
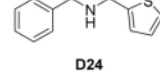
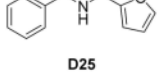
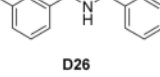
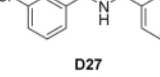
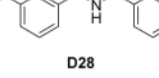
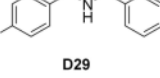
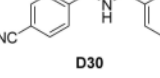
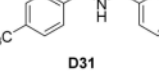
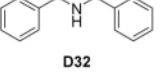
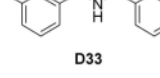
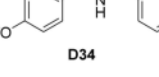
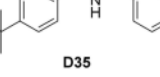
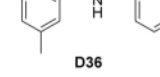
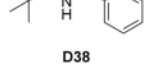
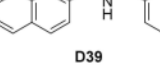
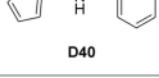
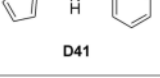
R_3 为H,或者 R_2 和 R_3 与N原子一起成五元环或六元环;

R_4 为H、 C_1-6 的烷基、一个或多个卤素取代的 C_1-6 的烷基、卤素、 C_1-6 的烷氧基或氰基;

R_5 为H、 C_1-6 的烷基、一个或多个卤素取代的 C_1-6 的烷基、卤素、 C_1-6 的烷氧基或氰基。

2. 如权利要求1所述的方法,其特征在于:所述“卤素”是指氟、氯、溴或碘;所述“ C_{16} 的烷基”选自甲基、乙基、丙基、异丙基、正丁基、异丁基、仲丁基、叔丁基、正戊基、异戊基、新戊基、正己基或异己基;所述“ C_{16} 的烷氧基”选自甲氧基、乙氧基或叔丁氧基。

3. 如权利要求1所述的方法,其特征在于,所述化合物D选自:

 D1	 D2	 D3
 D4	 D5	 D6
 D7	 D8	 D9
 D10	 D11	 D12
 D13	 D14	 D15
 D16	 D17	 D18
 D19	 D20	 D21
 D22	 D23	 D24
 D25	 D26	 D27
 D28	 D29	 D30
 D31	 D32	 D33
 D34	 D35	 D36
 D37	 D38	 D39
 D40	 D41	

4. 如权利要求1-3任一项所述的方法,其特征在于: α -羰基硫叶立德类化合物(A)、胺类化合物(B)和叠氮基三甲基硅烷(C)的摩尔比为1:1:1-3:1:3,以质量百分比计。

5. 如权利要求1-3任一项所述的方法,其特征在于:所述电化学合成方法使用的电解池为未分隔电解池;电极为石墨片、石墨毡、碳纸、网状玻璃碳、铂片、泡沫镍中的一种或多种。

6. 如权利要求5所述的方法,其特征在于:阳极为石墨片、石墨毡、碳纸、网状玻璃碳中的一种或几种,阴极为石墨片、石墨毡、碳纸、网状玻璃碳、铂片、泡沫镍中的一种或几种。

7. 如权利要求1-6任一项所述的方法,其特征在于:所述溶剂为N,N-二甲基甲酰胺、二甲基亚砆、N-甲基吡咯烷酮、乙腈中的一种。

8. 如权利要求1-7任一项所述的方法,其特征在于:反应时间为2-6 h。

9. 如权利要求1-8任一项所述的方法,其特征在于:电流为5-10 mA。

10. 如权利要求1所述的方法,其特征在于:所述电极石墨片为阳极,铂片为阴极;所述电解质为四正丁铵六氟磷酸盐,电解质浓度为0.1 M;电流为5 mA;溶剂为二甲基亚砆。

一种电化学合成酰胺类化合物的方法

技术领域

[0001] 本发明涉及有机合成技术领域,具体涉及一种电化学合成酰胺类化合物的方法。

背景技术

[0002] 酰胺键作为有机化学与生物化学中最关键、最普遍的官能团之一,赋予了酰胺类化合物广泛而重要的应用价值。这类化合物在化学、生命科学、材料科学及工业应用等多个领域中发挥着桥梁与纽带的核心作用。据统计,超过25%的药物分子中含有酰胺结构,凸显了其在药物化学中的特殊地位。从构成生命基础的蛋白质,到日常穿戴的尼龙纺织材料,再到疾病治疗药物,乃至工业生产与科学研究中的关键组分,酰胺类化合物几乎无处不在,兼具基础性与支柱性双重应用价值。

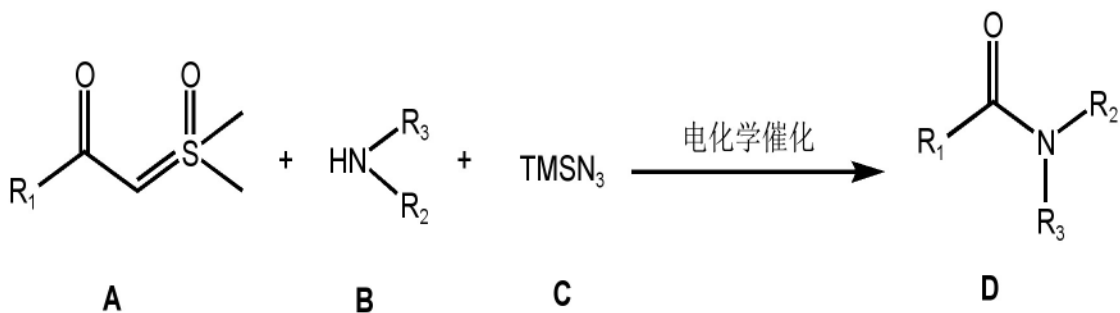
[0003] 目前,酰胺的合成主要采用以下几类方法:一、酸胺缩合法:羧酸与胺的缩合反应是构建酰胺键最常用的策略之一,通常借助缩合剂实现,如1-(3-二甲氨基丙基)-3-乙基碳二亚胺(EDCI)等碳二亚胺类缩合剂或鎓盐类缩合剂,该方法在实验室和工业规模中均有广泛应用。但碳二亚胺类缩合剂对于酸的 α -位存在空间位阻或连有吸电子基团的羧酸底物收率较低;而鎓盐类缩合剂存在价格高昂等局限性;二、酰卤法:酰氯或酰溴可与脂肪胺或芳香胺快速反应,高产率地生成酰胺。该类反应通常为剧烈放热过程;三、酸酐法:酸酐也可作为胺的酰化试剂,但其反应活性通常低于相应酰卤,因此与胺的反应速率较慢;四、酯的氨解:酯与氨水反应是实验室中制备酰胺的常用方法之一,但通常需在强碱或三甲基铝(AlMe_3)等试剂促进下才能顺利进行;五、重排与加成反应:包括Ritter反应、Beckmann重排、Schmidt重排及Overman重排等在内的多种重排与加成途径,也为酰胺的构建提供了重要手段,但这类反应往往对底物有特定要求,应用范围较为局限。

发明内容

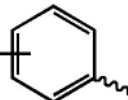
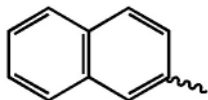
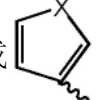
[0004] 现有酰胺合成方法通常需依赖缩合剂或催化剂,且存在反应条件苛刻、效率不高等问题。部分反应副产物难以分离,导致产物纯化困难。针对上述问题,本发明提出了一种电化学合成酰胺类化合物的新方法。该方法以 α -羰基硫叶立德、烷基胺和叠氮基三甲基硅烷为原料,通过电化学反应高效构建酰胺结构,生成目标酰胺产物D。

[0005] 为了实现上述目的,本发明的技术方案为:

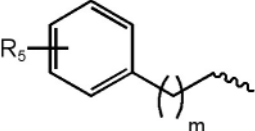
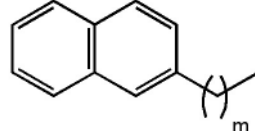
一种电化学合成酰胺类化合物D的方法,以化合物A,B,C为原料,反应式如下:

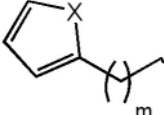


[0006] 其中,

R_1 选自 C_{1-6} 的烷基或环烷基、 R_4 -,  或 , 其中X为O或

S;

R_2 选自 C_{1-6} 的烷基或环烷基、 R_5 -,  或

, 其中X为O或S; m为1-3的整数;

R_3 为H, 或者 R_2 和 R_3 与N原子一起成五元环或六元环;

R_4 为H、 C_{1-6} 的烷基、一个或多个卤素取代的 C_{1-6} 的烷基、卤素、 C_{1-6} 的烷氧基或氰基;

R_5 为H、 C_{1-6} 的烷基、一个或多个卤素取代的 C_{1-6} 的烷基、卤素、 C_{1-6} 的烷氧基或氰基。

[0007] R_2 和 R_3 与N原子一起成五元环或六元环为四氢吡咯环、哌啶环或吗啉环。

[0008] 本文所用的术语“卤素”是指氟、氯、溴或碘, 优选为氟、氯或溴。

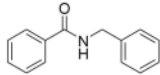
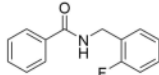
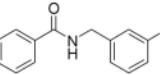
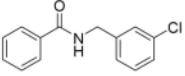
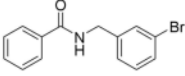
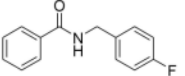
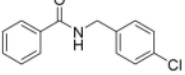
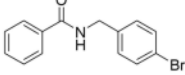
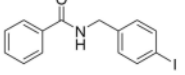
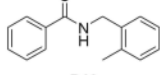
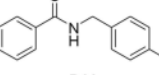
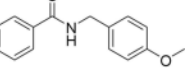
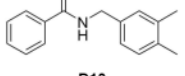
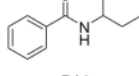
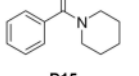
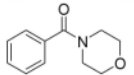
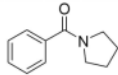
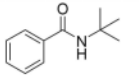
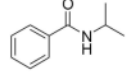
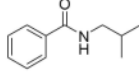
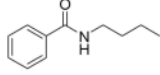
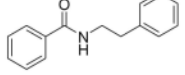
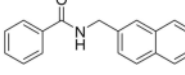
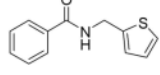
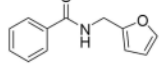
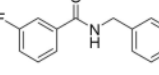
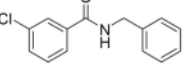
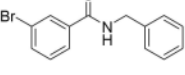
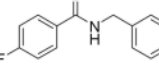
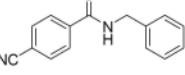
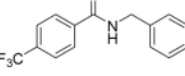
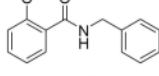
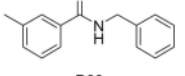
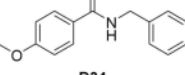
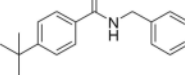
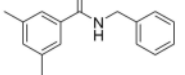
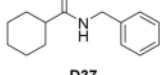
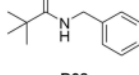
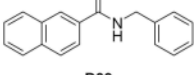
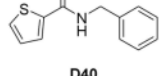
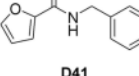
[0009] 本文所用的术语“ C_{16} 的烷基”是指具有16个碳原子的饱和的直链或支链烷基, 例如甲基、乙基、丙基、异丙基、正丁基、异丁基、仲丁基、叔丁基、正戊基、异戊基、新戊基、正己基、异己基等。本文所用的术语“ C_{16} 的环烷基为环己烷基”。

[0010] 本发明所用的术语“ C_{16} 的烷氧基”, 是指具有16个碳原子的饱和的直链或支链烷氧基, 例如甲氧基、乙氧基、叔丁氧基等。

[0011] 本发明所用的术语“一个或多个卤素取代的 C_{1-6} 的烷基”是指被多个卤素取代的直链或支链 C_{1-6} 的烷基, 比如三氟甲基、二氟甲基等。

[0012] 根据本发明的一个实施方案, R_1 为 C_{1-6} 烷基、苯基、卤代基苯基、甲基苯基、甲氧基苯基、叔丁基苯基、三氟甲基苯基、氰基苯基、萘基、噻吩基, 呋喃基中任一种。 R_2 为烷基、苄基、卤代基苄基、烷基苄基、甲氧基苄基、亚甲基噻吩基、亚甲基呋喃基、亚甲基萘基中任一种。

[0013] 根据本发明的一个实施方案, 所述化合物D选自:

 D1	 D2	 D3
 D4	 D5	 D6
 D7	 D8	 D9
 D10	 D11	 D12
 D13	 D14	 D15
 D16	 D17	 D18
 D19	 D20	 D21
 D22	 D23	 D24
 D25	 D26	 D27
 D28	 D29	 D30
 D31	 D32	 D33
 D34	 D35	 D36
 D37	 D38	 D39
 D40	 D41	

[0014] 根据本发明的一个实施方案, α -羰基硫叶立德类化合物(A)、胺类化合物(B)和叠氮基三甲基硅烷(C)的物质的量比例为1:1:1-3:1:3。

[0015] 根据本发明的一个实施方案,所述电化学合成方法使用的电解池为未分隔电解池;电极为石墨片、石墨毡、碳纸、网状玻璃碳、铂片、泡沫镍中的一种或多种。进一步,阳极为石墨片、石墨毡、碳纸、网状玻璃碳中的一种或几种,阴极为石墨片、石墨毡、碳纸、网状玻璃碳、铂片、泡沫镍中的一种或几种。

[0016] 根据本发明的一个实施方案,所述溶剂为N,N-二甲基甲酰胺、二甲基亚砜、N-甲基吡咯烷酮、乙腈中的一种。

[0017] 根据本发明的一个实施方案,反应时间为2-6 h。

[0018] 根据本发明的一个实施方案,电流为5-10 mA。

[0019] 根据本发明的一个实施方案,所述电极石墨片为阳极,铂片为阴极;所述电解质为四正丁铵六氟磷酸盐,电解质浓度为0.1 M;电流为5 mA;溶剂为二甲基亚砜。

[0020] 根据本发明的一个实施方案,所述反应温度为25 °C。

[0021] 根据本发明的一个实施方案,所述 α -羰基硫叶立德类化合物、胺类化合物和叠氮基三甲基硅烷的摩尔比为2:1:2,胺类化合物的浓度为0.05 M。

[0022] 根据本发明的一个实施方案,其特征在于,反应完成后,加水淬灭,用乙酸乙酯萃取三次,合并有机相,再用饱和氯化钠水溶液洗涤一次有机相,得有机相,减压浓缩溶剂,无需柱层析纯化即可得到产物。

[0023] 有益效果:

本发明提供了一种电化学合成酰胺类化合物的方法,通过高效的电化学策略, α -羰基硫叶立德、叠氮基三甲基硅烷和胺类化合物进行酰胺化,能以近乎当量收率得到酰胺化产物。本发明方法在温和条件下进行,无需高温条件,金属催化剂和化学氧化还原试剂,展现了环境友好性,具有高官能团耐受性,反应收率极高且纯化方式简单,并能应用于电化学连续流反应,适合工业化生产,具有极高产业化应用价值。

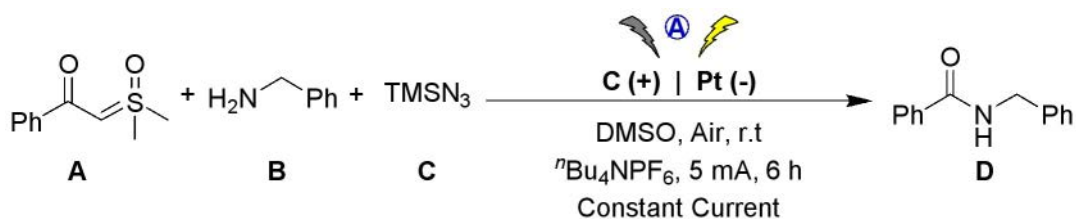
具体实施方式

[0024] 为使本发明实施例的目的、技术方案和优点更加清楚,下面将结合本申请实施例,对本申请实施例中的技术方案进行清楚、完整地描述,显然,所描述的实施例是本申请一部分实施例,而不是全部的实施例。如无特殊说明,本发明所记载的任何范围包括端值以及端值之间的任何数值以及端值或者端值之间的任意数值所构成的任意子范围。如无特殊说明,本发明所述百分比均为质量百分比。

[0025] 下面结合实例对本发明的内容进一步的阐述,但不是对本发明的限定。

[0026] 实施例1:

N-苄基苯甲酰胺D1的制备



[0027] 具体的操作步骤:将0.4 mmol的硫叶立德和0.4 mmol的四正丁铵六氟磷酸盐称于10 mL电解池中,电解池中设有1.5 cm×1.0 cm×0.1 cm的石墨片作为阳极和1.5 cm×1.0

cm×0.1 cm的铂片作为阴极,添加4 mL二甲基亚砷作为溶剂,加入0.2 mmol 苄胺和0.4 mmol 叠氮基三甲基硅烷,在室温下通电反应6 h,其中,通电电流为5 mA,采用TLC监测直至底物完全耗尽。反应完成后,加水淬灭,用乙酸乙酯萃取三次,合并有机相,再用饱和氯化钠水溶液洗涤一次有机相,得有机相,减压浓缩溶剂,得到产物。

[0028] 表1.不同底物比例对制备化合物D1的影响

序号	A:B:C	收率 (%)
1	1:1:2	56
2	2:1:2	91
3	3:1:2	91
4	3:1:3	99
5	1:2:2	50
6	2:1:3	92
7	3:1:1	58

[0029] 由表1可知,当底物比例为3:1:3的时候,产物的收率最高。

[0030] 表2.不同阳极材料对制备化合物D1的影响

序号	阳极材料	收率 (%)
1	石墨片	91
2	石墨毡	81
3	碳纸	84
4	网状玻璃碳	78

[0031] 由表2可知,当阳极材料为石墨片的时候,产物的收率最高。

[0032] 表3.不同阴极材料对制备化合物D1的影响

序号	阴极材料	收率 (%)
1	铂片	91
2	石墨片	75
3	石墨毡	73
4	碳纸	82
5	网状玻璃碳	74
6	泡沫镍	90

[0033] 由表3可知,当阴极材料为铂片的时候,产物的收率最高。

[0034] 表4.不同溶剂对制备化合物D1的影响

序号	溶剂	收率 (%)
1	DMSO	91
2	DMF	54
3	MeCN	24
4	NMP	51

[0035] 由表4可知,当溶剂为二甲基亚砩的时候,产物的收率最高。

[0036] 表5.不同时间对制备化合物D1的影响

序号	时间 (h)	收率 (%)
1	2	55
2	4	78
3	6	91

[0037] 由表5可知,当时间为6小时的时候,产物的收率最高。

[0038] 表6.不同电流(相同电荷量 $Q=I \times t$)对制备化合物D1的影响

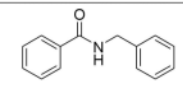
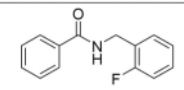
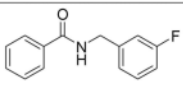
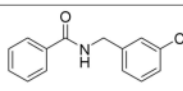
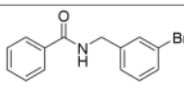
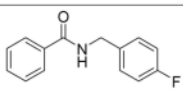
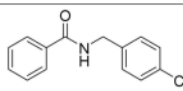
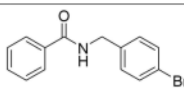
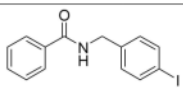
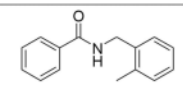
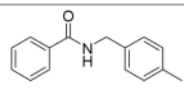
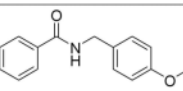
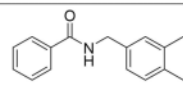
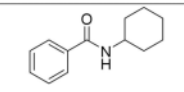
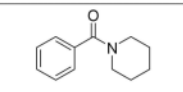
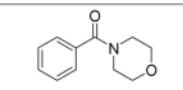
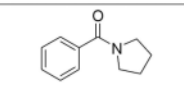
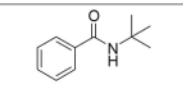
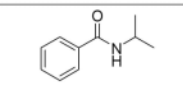
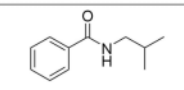
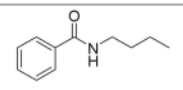
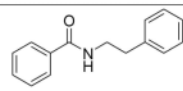
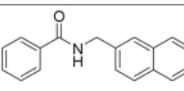
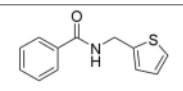
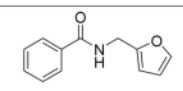
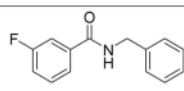
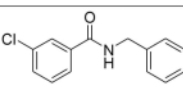
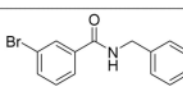
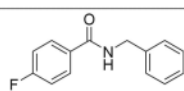
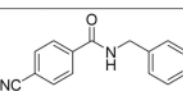
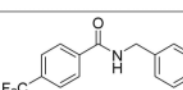
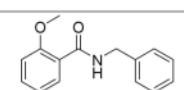
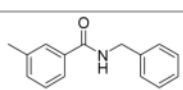
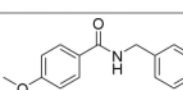
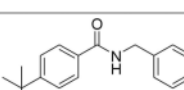
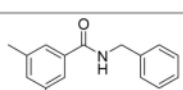
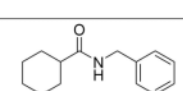
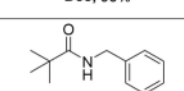
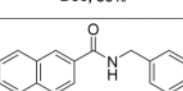
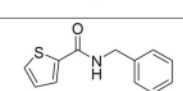
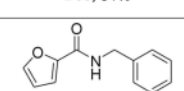
序号	电流大小	收率 (%)
1	5 mA, 6 h	91
2	10 mA, 3 h	63
3	0 mA, 6 h	N.R.

[0039] 由表6可知,当电流为5 mA的时候,反应6 h, 产物的收率最高。

[0040] 底物普适性的考察

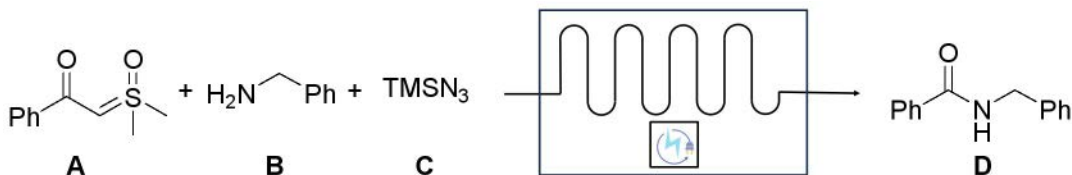
参照上述实施例最佳反应条件,制备下表D1-D41的化合物。

[0041]

 D1, 99%	 D2, 88%	 D3, 92%
 D4, 90%	 D5, 92%	 D6, 97%
 D7, 94%	 D8, 86%	 D9, 77%
 D10, 90%	 D11, 87%	 D12, 80%
 D13, 84%	 D14, 97%	 D15, 92%
 D16, 85%	 D17, 85%	 D18, 84%
 D19, 96%	 D20, 98%	 D21, 95%
 D22, 90%	 D23, 53%	 D24, 72%
 D25, 60%	 D26, 93%	 D27, 96%
 D28, 92%	 D29, 88%	 D30, 92%
 D31, 94%	 D32, 97%	 D33, 84%
 D34, 85%	 D35, 90%	 D36, 86%
 D37, 59%	 D38, 51%	 D39, 91%
 D40, 88%	 D41, 26%	

[0042]

电化学连续流反应实例：
N-苄基苯甲酰胺D1的制备

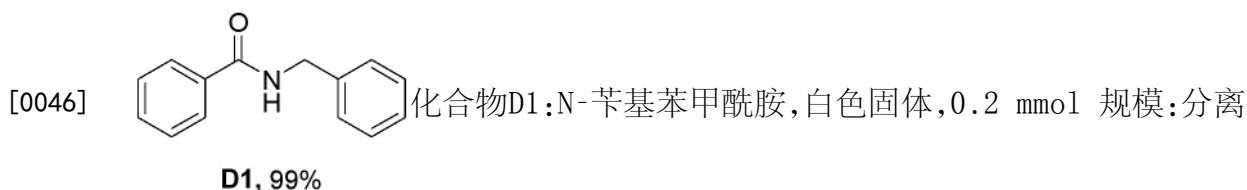


[0043] 具体的操作步骤:

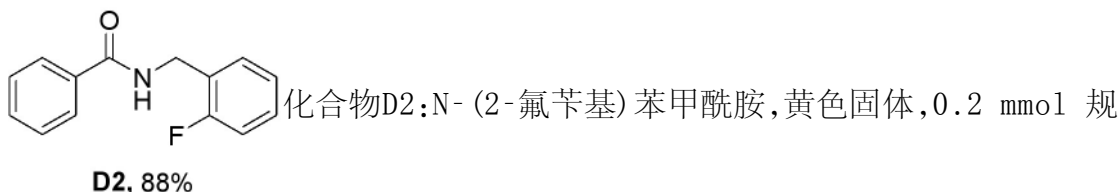
将溶于二甲亚砜中的硫叶立德底物(浓度 0.02 mol/L)、四正丁铵六氟磷酸盐(浓度 0.02 mol/L)、苄胺(浓度 0.01 mol/L)和叠氮基三甲基硅烷(浓度 0.02 mol/L)的混合溶液,在空气氛围下通过微量注射泵泵入流动电解池(石墨片作为阳极、铂片作为阴极)。控制流速为 0.3 毫升/分钟,并施加适宜电流,以确保物料单次通过电解池即可完全转化。收集 33.3 分钟的流出液(共计 10 毫升),经旋转蒸发仪减压浓缩。向浓缩后的残留物中加入 10 毫升水,随后用乙酸乙酯萃取三次(每次 10 毫升)。再用饱和氯化钠水溶液洗涤一次有机相,得有机相,减压浓缩溶剂,得到产物。

[0044] 综上,本发明提供了一种电化学合成酰胺类化合物的方法,通过高效的电化学策略, α -羰基硫叶立德、叠氮基三甲基硅烷和胺类化合物进行酰胺化,高效地合成了一系列酰胺化合物。本发明方法在温和条件下进行,无需缩合剂和催化剂,无明显副产物,表现出环境友好性且具有良好的官能团耐受性;反应收率高,纯化简单,无需柱层析操作,具有良好的合成实用性,适合工业化生产;且能应用于电化学连续流反应,具有极高的产业化应用价值。

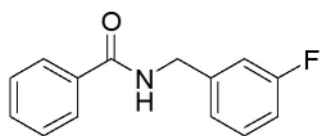
[0045] 化合物D1-D41的具体结构以及结构解析数据如下。



收率为99%(41.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.79 (d, $J = 7.2$ Hz, 2H), 7.49 (t, $J = 7.2$ Hz, 1H), 7.41 (t, $J = 7.2$ Hz, 2H), 7.32 (m, 5H), 6.63 (br s, 1H), 4.62 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.5, 138.3, 134.5, 131.6, 128.9, 128.7, 128.0, 127.7, 127.1, 44.2.



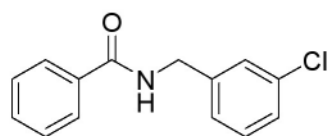
模:分离收率为88%(40.3 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.81-7.72 (m, 2H), 7.53-7.43 (m, 1H), 7.41-7.33 (m, 3H), 7.29-7.19 (m, 1H), 7.12-6.98 (m, 2H), 6.75 (br s, 1H), 4.65 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.6, 161.1 (d, $J = 246.9$ Hz), 134.3, 131.7, 130.4 (d, $J = 4.3$ Hz), 129.4 (d, $J = 8.2$ Hz), 128.6, 127.1, 125.3 (d, $J = 14.8$ Hz), 124.4 (d, $J = 3.4$ Hz), 115.5 (d, $J = 21.3$ Hz), 38.1 (d, $J = 3.9$ Hz). ^{19}F NMR (376 MHz, CDCl_3) δ -118.8.



化合物D3:N-(3-氟苄基)苯甲酰胺,黄色固体,0.2 mmol

D3, 92%

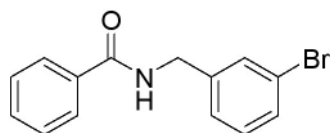
规模:分离收率为92%(45.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.78 (d, $J = 7.2$ Hz, 2H), 7.47 (t, $J = 7.6$ Hz, 1H), 7.38 (t, $J = 7.6$ Hz, 2H), 7.30-7.20 (m, 1H), 7.06 (d, $J = 7.6$ Hz, 1H), 6.99 (d, $J = 10$ Hz, 1H), 6.96-6.89 (m, 2H), 4.56 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 163.1 (d, $J = 247.3$ Hz), 141.0 (d, $J = 7.3$ Hz), 134.2, 131.8, 130.3 (d, $J = 8.2$ Hz), 128.7, 127.1, 123.3 (d, $J = 2.9$ Hz), 114.5 (d, $J = 40.7$ Hz), 114.5 (d, $J = 2.4$ Hz), 43.5. ^{19}F NMR (376 MHz, CDCl_3) δ -112.6.



化合物D4:N-(3-氯苄基)苯甲酰胺,黄色固体,0.2 mmol

D4, 90%

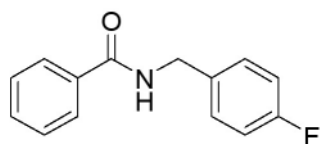
规模:分离收率为90%(44.1 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.79 (d, $J = 7.2$ Hz, 2H), 7.49 (t, $J = 7.2$ Hz, 1H), 7.39 (t, $J = 8.0$ Hz, 2H), 7.29 (s, 1H), 7.25-7.14 (m, 3H), 6.92 (br s, 1H), 4.55 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 140.5, 134.6, 134.1, 131.8, 130.1, 128.7, 127.8, 127.7, 127.1, 125.9, 43.5.



化合物D5:N-(3-溴苄基)苯甲酰胺,黄色固体,0.2 mmol

D5, 92%

规模:分离收率为92%(53.2 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.82-7.73 (m, 2H), 7.54-7.43 (m, 2H), 7.42-7.36 (m, 3H), 7.23 (s, 1H), 7.17 (t, $J = 7.6$ Hz, 1H), 6.86 (br s, 1H), 4.56 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 140.8, 134.1, 131.8, 130.8, 130.7, 130.4, 128.7, 127.2, 126.5, 122.8, 43.5.

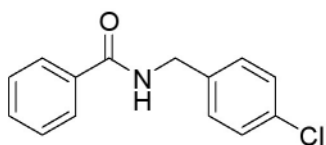


化合物D6:N-(4-氟苄基)苯甲酰胺,黄色固体,0.2 mmol

D6, 97%

规模:分离收率为97%(44.4 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.88-7.69 (m, 2H), 7.53-7.44 (m, 1H), 7.39 (t, $J = 7.6$ Hz, 2H), 7.32-7.22 (m, 2H), 6.98 (t, $J = 8.4$ Hz, 2H), 6.83 (br s, 1H), 4.55 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.6, 162.3 (d, $J = 246.8$ Hz), 134.3, 134.2 (d, $J = 3.3$ Hz), 131.7, 129.6 (d, $J = 8.2$ Hz), 128.7, 127.1, 115.6 (d, $J = 21.7$ Hz), 43.4. ^{19}F NMR (376 MHz,

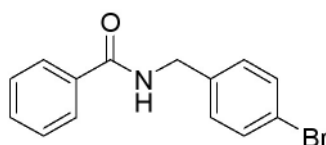
CDCl_3) δ -114.9.



化合物D7:N-(4-氯苄基)苯甲酰胺,黄色固体,0.2 mmol

D7, 94%

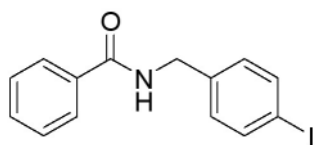
规模:分离收率为94%(46.1 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.82-7.73 (m, 2H), 7.53-7.44 (m, 1H), 7.38 (m, 2H), 7.31-7.17 (m, 4H), 6.82 (br s, 1H), 4.53 (d, J = 5.6 Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.6, 136.9, 134.2, 133.4, 131.8, 129.3, 128.9, 128.7, 127.1, 43.4.



化合物D8:N-(4-溴苄基)苯甲酰胺,黄色固体,0.2 mmol

D8, 86%

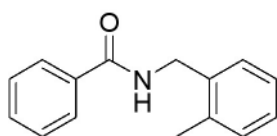
规模:分离收率为86%(49.7 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.77 (d, J = 7.2 Hz, 2H), 7.54-7.44 (m, 1H), 7.46-7.34 (m, 4H), 7.17 (d, J = 8.4 Hz, 2H), 6.88 (br s, 1H), 4.52 (d, J = 6.0 Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 137.5, 134.2, 131.9, 131.8, 129.6, 128.7, 127.1, 121.5, 43.4.



化合物D9:N-(4-碘苄基)苯甲酰胺,黄色固体,0.2 mmol

D9, 77%

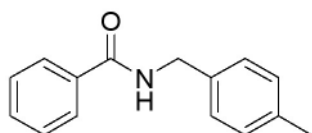
规模:分离收率为77%(51.9 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.77 (d, J = 6.8 Hz, 2H), 7.61 (d, J = 8.4 Hz, 2H), 7.48 (t, J = 7.6 Hz, 1H), 7.38 (t, J = 7.6 Hz, 2H), 7.03 (d, J = 8.4 Hz, 2H), 6.92 (br s, 1H), 4.50 (d, J = 6.0 Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.6, 138.2, 137.8, 134.2, 131.8, 129.8, 128.7, 127.1, 93.0, 43.5.



化合物D10:N-(2-甲基苄基)苯甲酰胺,黄色固体,0.2 mmol

D10, 90%

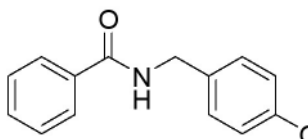
规模:分离收率为90%(40.5 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.82-7.73 (m, 2H), 7.54-7.45 (m, 1H), 7.40 (t, J = 8.0 Hz, 2H), 7.29 (d, J = 6.8 Hz, 1H), 7.23-7.15 (m, 3H), 6.46 (br s, 1H), 4.61 (d, J = 5.6 Hz, 2H), 2.36 (s, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.4, 136.6, 135.9, 134.4, 131.6, 130.7, 128.7, 128.6, 127.9, 127.1, 126.3, 42.4, 19.1.



化合物D11:N-(4-甲基苄基)苯甲酰胺,黄色固体,0.2 mmol

D11, 87%

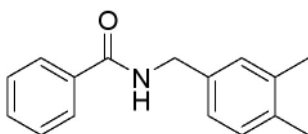
规模:分离收率为87%(39.2 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.82-7.75 (m, 2H), 7.53-7.44 (m, 1H), 7.40 (t, $J = 8.4$ Hz, 2H), 7.23 (d, $J = 8.0$ Hz, 2H), 7.15 (d, $J = 8.0$ Hz, 2H), 6.65 (br s, 1H), 4.57 (d, $J = 5.6$ Hz, 2H), 2.34 (s, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.5, 137.3, 135.3, 134.5, 131.5, 129.5, 128.6, 127.9, 127.1, 43.9, 21.2.



化合物D12:N-(4-甲氧基苄基)苯甲酰胺,黄色固体,0.2

D12, 80%

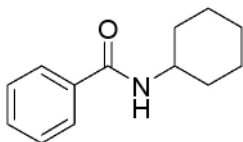
mmol 规模:分离收率为80%(38.6 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.81-7.73 (m, 2H), 7.52-7.43 (m, 1H), 7.42-7.36 (m, 2H), 7.26 (d, $J = 8.8$ Hz, 2H), 6.86 (d, $J = 8.4$ Hz, 2H), 6.57 (br s, 1H), 4.55 (d, $J = 5.6$ Hz, 2H), 3.78 (s, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.4, 159.2, 134.5, 131.6, 130.4, 129.4, 128.7, 127.1, 114.2, 55.4, 43.7.



化合物D13:N-(3,4-二甲基苄基)苯甲酰胺,黄色固体,0.2

D13, 84%

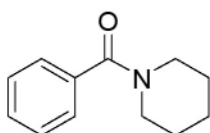
mmol 规模:分离收率为84%(40.2 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.79 (d, $J = 6.8$ Hz, 2H), 7.49 (t, $J = 7.2$ Hz, 1H), 7.45-7.36 (m, 2H), 7.15-6.99 (m, 3H), 6.57 (br s, 1H), 4.56 (d, $J = 5.6$ Hz, 2H), 2.25 (s, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.4, 137.1, 136.0, 135.7, 134.6, 131.5, 130.0, 129.4, 128.6, 127.1, 125.5, 44.0, 19.8, 19.5.



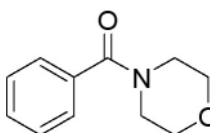
化合物D14:N-环己基苯甲酰胺,黄色固体,0.2 mmol 规模:分离

D14, 97%

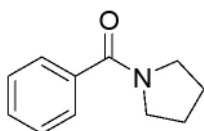
收率为97%(39.4 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.77-7.71 (m, 2H), 7.50-7.43 (m, 1H), 7.42-7.36 (m, 2H), 6.11 (br s, 1H), 3.95 (m, 1H), 2.01 (m, 2H), 1.78-1.69 (m, 2H), 1.68-1.57 (m, 1H), 1.48-1.30 (m, 2H), 1.30-1.10 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 166.8, 135.2, 131.3, 128.6, 127.0, 48.8, 33.3, 25.6, 25.0.

**D15, 92%**

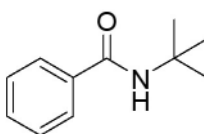
化合物D15:1-苯甲酰哌啶,黄色油状,0.2 mmol 规模:分离收率为92%(34.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.37 (s, 5H), 3.69 (s, 2H), 3.32 (s, 2H), 1.66 (s, 4H), 1.50 (s, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 170.4, 136.6, 129.4, 128.5, 126.9, 48.8, 43.2, 26.6, 25.7, 24.7.

**D16, 85%**

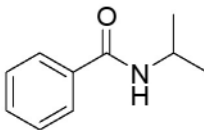
化合物D16:吗啉基(苯基)甲酮,黄色油状,0.2 mmol 规模:分离收率为85%(32.5 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.46-7.34 (m, 5H), 3.94-3.27 (m, 8H). ^{13}C NMR (101 MHz, CDCl_3) δ 170.5, 135.4, 130.0, 128.7, 127.2, 67.0, 48.3, 42.6.

**D17, 85%**

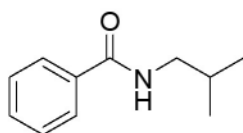
化合物D17:苯基(吡咯烷-1-基)甲酮,黄色油状,0.2 mmol 规模:分离收率为85%(29.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.52-7.46 (m, 2H), 7.37 (d, J = 5.6 Hz, 3H), 3.63 (t, J = 6.8 Hz, 2H), 3.40 (t, J = 6.8 Hz, 2H), 1.98-1.89 (m, 2H), 1.89-1.80 (m, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 169.8, 137.3, 129.8, 128.3, 127.1, 49.7, 46.2, 26.5, 24.5.

**D18, 84%**

化合物D18:N-叔丁基苯甲酰胺,白色固体,0.2 mmol 规模:分离收率为84%(29.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.76-7.67 (m, 2H), 7.53-7.42 (m, 1H), 7.43-7.37 (m, 2H), 5.95 (br s, 1H), 1.47 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.0, 136.0, 131.2, 128.6, 126.8, 51.7, 29.0.

**D19, 96%**

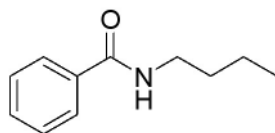
化合物D19:N-异丙基苯甲酰胺,黄色固体,0.2 mmol 规模:分离收率为96%(31.3 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.74 (d, J = 6.8 Hz, 2H), 7.49-7.43 (m, 1H), 7.42-7.36 (m, 2H), 6.06 (br s, 1H), 4.36-4.19 (m, 1H), 1.25 (d, J = 6.4 Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 166.8, 135.1, 131.3, 128.6, 126.9, 42.0, 22.9.



化合物D20:N-异丁基苯甲酰胺,黄色固体,0.2 mmol 规模:分离

D20, 98%

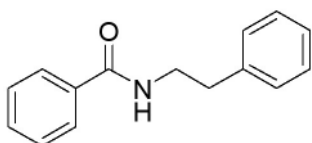
收率为98%(34.7 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.80-7.71 (m, 2H), 7.53-7.43 (m, 1H), 7.42-7.36 (m, 2H), 6.44 (br s, 1H), 3.33-3.15 (m, 2H), 1.88 (m, 1H), 0.95 (d, $J = 6.4$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.8, 135.0, 131.4, 128.6, 127.0, 47.4, 28.7, 20.3.



化合物D21:N-丁基苯甲酰胺,黄色油状,0.2 mmol 规模:分离

D21, 95%

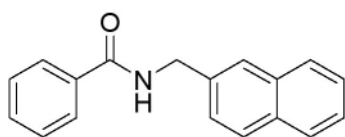
收率为95%(33.7 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.75 (d, $J = 7.2$ Hz, 2H), 7.48-7.43 (m, 1H), 7.42-7.35 (m, 2H), 6.42 (br s, 1H), 3.47-3.34 (m, 2H), 1.62-1.52 (m, 2H), 1.44-1.32 (m, 2H), 0.93 (t, $J = 7.2$ Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 134.9, 131.3, 128.6, 127.0, 39.9, 31.8, 20.2, 13.9.



化合物D22:N-苯乙基苯甲酰胺,黄色固体,0.2 mmol 规模:

D22, 90%

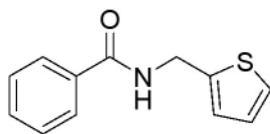
分离收率为90%(40.5 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.71-7.67 (m, 2H), 7.51-7.43 (m, 1H), 7.41-7.34 (m, 2H), 7.34-7.27 (m, 2H), 7.26-7.19 (m, 3H), 6.38 (br s, 1H), 3.74-3.63 (m, 2H), 2.92 (t, $J = 7.2$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 139.1, 134.8, 131.5, 128.9, 128.8, 128.7, 127.0, 126.7, 41.3, 35.8.



化合物D23:N-(萘-2-基甲基)苯甲酰胺,棕色固体,0.2

D23, 53%

mmol 规模:分离收率为53%(27.7 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.85-7.74 (m, 6H), 7.54-7.35 (m, 6H), 6.67 (br s, 1H), 4.78 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.6, 135.8, 134.5, 133.5, 132.9, 131.7, 128.8, 127.9, 127.8, 127.1, 126.6, 126.4, 126.1, 44.4.

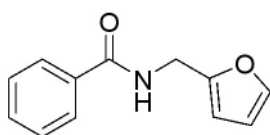


化合物D24:N-(噻吩-2-基甲基)苯甲酰胺,棕色固体,0.2

D24, 72%

mmol 规模:分离收率为72%(31.3 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.81-7.75 (m, 2H),

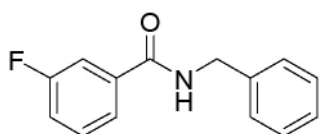
7.53-7.45 (m, 1H), 7.45-7.36 (m, 2H), 7.28-7.20 (m, 1H), 7.06-6.99 (m, 1H), 6.98-6.93 (m, 1H), 6.64 (br s, 1H), 4.79 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.3, 140.9, 134.3, 131.7, 128.7, 127.1, 127.1, 126.3, 125.4, 38.9.



化合物D25:N-(呋喃-2-基甲基)苯甲酰胺,棕色固体,0.2

D25, 60%

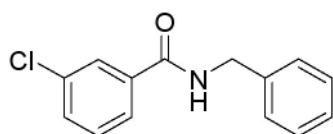
mmol 规模:分离收率为60%(24.1 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.78 (d, $J = 6.8$ Hz, 2H), 7.50 (t, $J = 7.6$ Hz, 1H), 7.42 (t, $J = 7.6$ Hz, 2H), 7.38 (s, 1H), 6.47 (br s, 1H), 6.32 (d, $J = 14.4$ Hz, 2H), 4.64 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.4, 151.3, 142.5, 134.3, 131.8, 128.7, 127.1, 110.7, 107.9, 37.2.



化合物D26:N-苄基-3-氟苯甲酰胺,黄色固体,0.2 mmol

D26, 93%

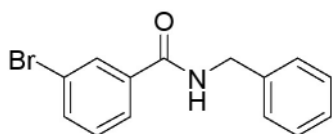
规模:分离收率为93%(42.6 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.56-7.47 (m, 2H), 7.42-7.26 (m, 6H), 7.23-7.13 (m, 1H), 6.63 (br s, 1H), 4.60 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 166.3 (d, $J = 2.7$ Hz), 162.8 (d, $J = 248.8$ Hz), 138.0, 136.7 (d, $J = 6.8$ Hz), 130.3 (d, $J = 14.0$ Hz), 128.8, 128.0, 127.9, 122.6 (d, $J = 3.1$ Hz), 118.5 (d, $J = 18.3$ Hz), 114.6 (d, $J = 16.5$ Hz), 44.2. ^{19}F NMR (376 MHz, CDCl_3) δ -111.7.



化合物D27:N-苄基-3-氯苯甲酰胺,黄色固体,0.2 mmol

D27, 96%

规模:分离收率为96%(47.1 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.77 (t, $J = 2$ Hz, 1H), 7.64 (d, $J = 7.6$ Hz, 1H), 7.45 (d, $J = 8.0$ Hz, 1H), 7.40-7.24 (m, 6H), 6.73 (br s, 1H), 4.59 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 166.3, 138.0, 136.2, 134.8, 131.6, 130.0, 128.9, 128.0, 127.8, 127.5, 125.2, 44.3.

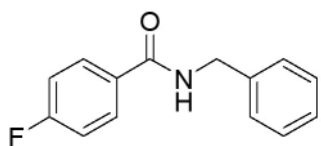


化合物D28:N-苄基-3-溴苯甲酰胺,白色固体,0.2 mmol

D28, 92%

规模:分离收率为92%(53.2 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.91 (t, $J = 2.0$ Hz, 1H), 7.68 (d, $J = 8.0$ Hz, 1H), 7.59 (d, $J = 8.0$ Hz, 1H), 7.40-7.18 (m, 6H),

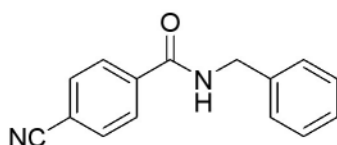
6.68 (br s, 1H), 4.58 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 166.2, 138.0, 136.4, 134.6, 130.4, 130.2, 128.9, 128.0, 127.8, 125.7, 122.8, 44.3.



D29, 88%

化合物D29:N-苄基-4-氟苯甲酰胺,黄色固体,0.2 mmol

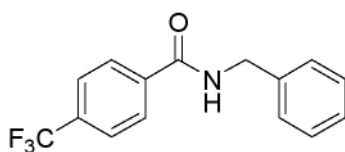
规模:分离收率为88%(40.3 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.84-7.74 (m, 2H), 7.36-7.27 (m, 5H), 7.07 (t, $J = 8.8$ Hz, 2H), 6.63 (br s, 1H), 4.60 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 166.6, 164.8 (d, $J = 252.9$ Hz), 138.2, 130.6 (d, $J = 2.9$ Hz), 129.5 (d, $J = 8.9$ Hz), 128.9, 127.9, 127.7, 115.6 (d, $J = 21.7$ Hz), 44.2. ^{19}F NMR (376 MHz, CDCl_3) δ -108.0.



D30, 92%

化合物D30:N-苄基-4-氰基苯甲酰胺,黄色固体,0.2

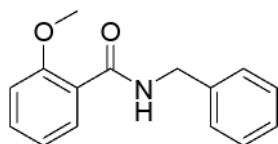
mmol 规模:分离收率为92%(43.4 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.85 (d, $J = 8.0$ Hz, 2H), 7.64 (d, $J = 6.8$ Hz, 2H), 7.38-7.24 (m, 5H), 7.04 (br s, 1H), 4.58 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 165.8, 138.3, 137.7, 132.5, 128.9, 127.9, 127.9, 127.1, 118.1, 115.1, 44.4.



D31, 94%

化合物D31:N-(4-碘苄基)苯甲酰胺,黄色固体,0.2

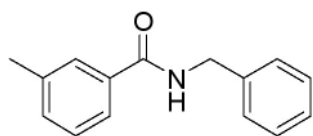
mmol 规模:分离收率为94%(52.5 mg), ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 9.34 (t, $J = 6.0$ Hz, 1H), 8.11 (d, $J = 8.0$ Hz, 2H), 7.85 (d, $J = 8.4$ Hz, 2H), 7.37-7.30 (m, 4H), 7.28-7.21 (m, 1H), 4.51 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ 165.1, 139.4, 138.1, 131.2 (q, $J = 32.1$ Hz), 128.3, 128.2, 127.3, 126.9, 125.4, 124.0 (q, $J = 272.0$ Hz), 42.8. ^{19}F NMR (376 MHz, $\text{DMSO}-d_6$) δ -61.3.



D32, 97%

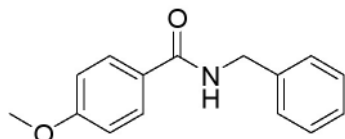
化合物D32:N-苄基-2-甲氧基苯甲酰胺,黄色油状,0.2 mmol

规模:分离收率为97%(46.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 8.27-8.22 (m, 1H), 8.19 (br s, 1H), 7.48-7.39 (m, 1H), 7.38-7.30 (m, 4H), 7.29-7.22 (m, 1H), 7.08 (t, $J = 7.6$ Hz, 1H), 6.95 (d, $J = 8.4$ Hz, 1H), 4.68 (d, $J = 5.6$ Hz, 2H), 3.89 (s, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 165.4, 157.6, 138.9, 132.9, 132.4, 128.7, 127.6, 127.3, 121.5, 121.4, 111.4, 56.0, 43.8.

**D33, 84%**

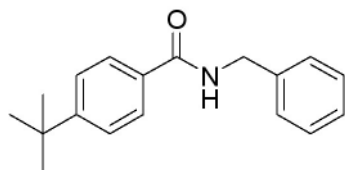
化合物D33:N-苄基-3-甲基苯甲酰胺,黄色固体,0.2 mmol

规模:分离收率为84%(37.8 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.62 (s, 1H), 7.59-7.54 (m, 1H), 7.34 (d, $J = 4.8$ Hz, 4H), 7.32-7.28 (m, 3H), 6.59 (br s, 1H), 4.62 (d, $J = 6.0$ Hz, 2H), 2.37 (s, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.7, 138.5, 138.4, 134.4, 132.3, 128.8, 128.5, 128.0, 127.9, 127.6, 124.0, 44.1, 21.4.

**D34, 85%**

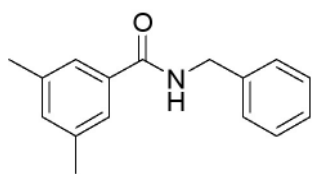
化合物D34:N-苄基-4-甲氧基苯甲酰胺,黄色固体,0.2

mmol 规模:分离收率为85%(41.0 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.75 (d, $J = 8.8$ Hz, 2H), 7.36-7.23 (m, 5H), 6.88 (d, $J = 8.8$ Hz, 2H), 6.58 (br s, 1H), 4.59 (d, $J = 5.6$ Hz, 2H), 3.82 (s, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.1, 162.3, 138.6, 128.9, 128.8, 128.0, 127.6, 126.7, 113.8, 55.5, 44.1.

**D35, 90%**

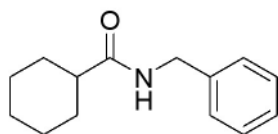
化合物D35:N-苄基-4-叔丁基苯甲酰胺,黄色固体,0.2

mmol 规模:分离收率为90%(48.1 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.74 (d, $J = 8.8$ Hz, 2H), 7.43 (d, $J = 8.4$ Hz, 2H), 7.35-7.27 (m, 5H), 6.65 (br s, 1H), 4.62 (d, $J = 5.6$ Hz, 2H), 1.33 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.5, 155.1, 138.5, 131.6, 128.8, 127.9, 127.6, 127.0, 125.6, 44.1, 35.0, 31.3.

**D36, 86%**

化合物D36:N-苄基-3,5-二甲基苯甲酰胺,黄色固体,0.2

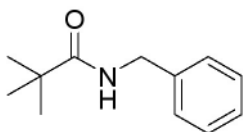
mmol 规模:分离收率为86%(41.1 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.40 (s, 2H), 7.35 (d, $J = 4.8$ Hz, 4H), 7.32-7.27 (m, 1H), 7.15-7.09 (m, 1H), 6.52 (br s, 1H), 4.62 (d, $J = 5.6$ Hz, 2H), 2.33 (s, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.9, 138.5, 138.3, 134.5, 133.2, 128.8, 128.0, 127.6, 124.9, 44.1, 21.3.

**D37, 59%**

化合物D37:N-苄基环己烷甲酰胺,黄色固体,0.2 mmol 规模:

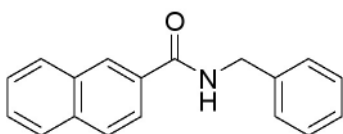
分离收率为59%(25.6 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.35-7.30 (m, 2H), 7.29-7.23

(m, 3H), 5.82 (br s, 1H), 4.42 (d, $J = 6.0$ Hz, 2H), 2.17-2.05 (m, 1H), 1.88 (d, $J = 13.6$ Hz, 2H), 1.83-1.73 (m, 2H), 1.71-1.62 (m, 1H), 1.54-1.38 (m, 2H), 1.34-1.13 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 176.1, 138.6, 128.8, 127.8, 127.6, 45.7, 43.5, 29.8, 25.8.



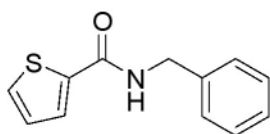
D38, 51%

率为51% (19.5 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.36-7.31 (m, 2H), 7.30-7.22 (m, 3H), 5.93 (br s, 1H), 4.43 (d, $J = 5.6$ Hz, 2H), 1.23 (s, 9H). ^{13}C NMR (101 MHz, CDCl_3) δ 178.5, 138.8, 128.9, 127.8, 127.6, 43.7, 38.9, 27.7.



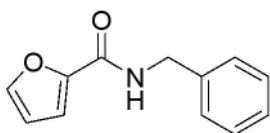
D39, 91%

模:分离收率为91% (47.5 mg), ^1H NMR (400 MHz, CDCl_3) δ 8.30 (s, 1H), 8.01-7.74 (m, 4H), 7.59-7.43 (m, 2H), 7.40-7.25 (m, 5H), 6.84 (br s, 1H), 4.68 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 167.6, 138.3, 134.9, 132.7, 131.7, 129.0, 128.9, 128.6, 128.0, 127.9, 127.8, 127.6, 127.6, 126.9, 123.7, 44.4.



D40, 88%

模:分离收率为88% (38.2 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.53 (dd, $J = 3.6, 1.2$ Hz, 1H), 7.45 (dd, $J = 4.8, 1.2$ Hz, 1H), 7.35-7.22 (m, 5H), 7.12-6.97 (m, 1H), 6.63 (br s, 1H), 4.57 (d, $J = 5.6$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 162.0, 138.9, 138.2, 130.2, 128.9, 128.3, 128.0, 127.8, 127.7, 44.1.



D41, 26%

模:分离收率为26% (10.5 mg), ^1H NMR (400 MHz, CDCl_3) δ 7.42 (dd, $J = 2.0, 1.2$ Hz, 1H), 7.35 (d, $J = 4.4$ Hz, 4H), 7.34-7.25 (m, 1H), 7.15 (dd, $J = 3.2, 0.8$ Hz, 1H), 6.64 (br s, 1H), 6.50 (q, $J = 1.6$ Hz, 1H), 4.62 (d, $J = 6.0$ Hz, 2H). ^{13}C NMR (101 MHz, CDCl_3) δ 158.4, 148.0, 144.0, 138.1, 128.9, 128.1, 127.8, 114.6, 112.3, 43.3.